

CrackID: A Within-Visit Benchmark for Crack Re-Identification, and the Limits of Appearance Matching

Cátia Loureiro, António Silva, Beatriz Miranda, and Paulo Brito
Délcio Ferramenta, Nuno da Costa, Ricardo Rodrigues, and Duarte Fernandes
DTx Colab, Campus de Azurém, Universidade do Minho, 4800-058 Braga, Portugal

Abstract—Maintenance systems turn photographs into tracked defect records: a crack re-photographed later must be matched to its record or filed as new. Crack *detection* is mature, but no benchmark for crack *re-identification* is known to us. We introduce CRACKID: 140 photographs of 17 walls with 212 crack identities, captured in one continuous walk-around per wall, so it measures within-visit viewpoint and scale robustness, not cross-visit re-identification. Every metric is reported against its chance level, with an annotation-quality report (45% of identities from an automatic merge). We benchmark ten crop-based matchers from four families. Ranking metrics separate them (Rank-1 0.352–0.654; chance 0.336), but on the thresholded same-crack decision none exceeds the 0.305 pairwise- F_1 floor by more than 0.054. Two controls localize the difficulty: adding wall context raises OSNet’s open-set rate from 0.168 to 0.404, yet inpainting the crack out of that view retains 0.393; context methods retrieve on location, not crack appearance. A geometric reference baseline that registers on wall texture with cracks masked, then Chamfer-matches projected crack masks, reaches $\text{DIR@FAR}=0.1$ of 0.768 against 0.429 for the best crop method; a coverage-only control scores 0.000. It answers only 48% of pairs, fails on two smooth-plaster walls, and is entangled with the homography-based provenance of our annotations; on a synthetic revisit check, OSNet+context outperforms it in ranking and open-set rate. All results are point estimates on 19 identities across 11 walls; we claim the pattern, not the digits.

Index Terms—crack re-identification, dataset, benchmark, building maintenance, structural inspection, homography, geometric alignment, annotation quality

I. INTRODUCTION

Automated building-maintenance systems convert photographs into tracked defect records: a re-photographed defect must eventually be *re-identified*, matched to its record, or given a new one. This paper first studies within-visit re-identification under viewpoint change. Wall cracks are a common and difficult case, because the same crack is re-photographed from different distances, angles, and lighting, and (Sec. III) grows.

Crack analysis is a mature *detection* and *segmentation* task, and its datasets, CrackSeg9k [1] and OmniCrack30k [2], carry no cross-observation identity labels, because the task they serve needs none. We know of no benchmark for crack *re-identification*; the natural approach, crop-and-match as in person and vehicle re-ID [3], [4], assumes a crack has stable, locally distinctive appearance, and whether it does has never been measured. We introduce CRACKID, 140 photographs, 17 walls, 212 crack identities, to make that re-identification mea-

surement (Fig. 1). Capture is one continuous walk-around per wall, so what is measured is *within-visit* viewpoint and scale correspondence, an upper bound on multi-visit performance; we add a synthetic revisit protocol as a controlled secondary check (Sec. VII-D). That check uses a distinct *Hybrid* pipeline: it extends the Table II REG+CHAMFER registration baseline with a skeleton-comparison fallback and per-pair blending, so its reversal does not directly contradict Table II.

Appearance buys almost nothing on the decision. On ranking the ten matchers separate as expected (Rank-1 0.352 to 0.654 against 0.336 chance); on pairwise F_1 , the *same-crack-or-not* decision a record makes at a threshold, all ten land between 0.000 and 0.054 above the 0.305 floor, three (ORB, DeiT, DINOv2) at exactly zero, and at a validation-calibrated threshold nine of the ten span only -0.012 to $+0.012$ around chance while LoFTR holds a real margin of 0.046.

Two controls say where the difficulty is not. Giving an embedder the surrounding wall raises OSNet’s open-set rate from 0.168 to 0.404; *erasing the crack* from that view costs almost none of it (0.393). Such a method retrieves on the wall, and the cue that works, even implicitly, is *location*.

A geometric baseline makes location explicit. Registering on non-crack texture, projecting crack masks into a common frame, and Chamfer-matching under global assignment reaches $\text{DIR@FAR}=0.1$ of 0.768 against 0.429 (best crop) and 0.404 (best context), answering 48% of pairs; a COVERAGE-ONLY control scores 0.000 on the same metric, so the open-set gap is matching, not alignment coverage. This is a conditional result, not a deployment-ready system: two test walls have zero registration success, and the annotation and baseline both use homography-based alignment. The resulting geometric advantage is therefore a hypothesis about this benchmark, not an independently established identity effect.

The revisit check reverses the ranking story. The real data are one continuous walk-around per wall, minutes apart, not separate visits. Under the synthetic warp-plus-appearance protocol, OSNet+context beats the hybrid at R@1 (0.826/0.671 versus 0.275/0.274) and open-set rate (0.739/0.663 versus 0.343/0.342) for growth/appearance change; the hybrid’s small assignment- F_1 advantage is not evidence that geometry solves real multi-visit re-identification.

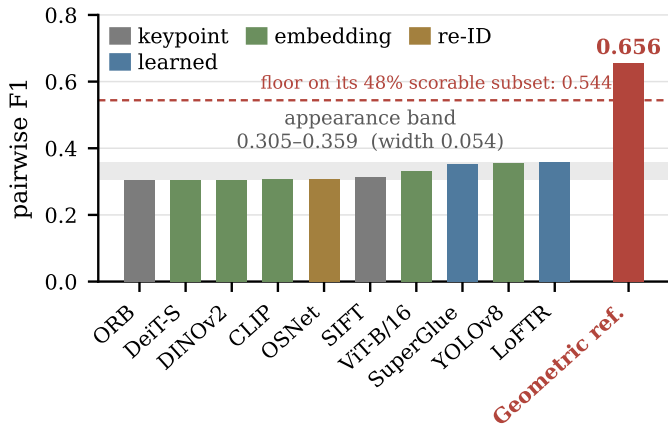


Fig. 1. **The appearance ceiling.** Pairwise F_1 for ten crop-scope matchers spanning four method families on CRACKID test. Every method falls inside the band 0.305 to 0.359, width 0.054, pressed against the 0.305 chance floor ($2p/(1+p)$ for this pool), irrespective of family, era, or compute budget. The geometric reference baseline (red) leaves the band, but it answers only 48% of pairs, for which the best-constant floor is 0.544 (the COVERAGE-ONLY level), not 0.305: its bar is drawn against the dashed line. Ranking metrics separate the same methods over a 0.30 range; the decision metric does not.

Contributions: (i) CRACKID with an explicit annotation-quality report, 45% of identities were formed by an automatic merge, reported rather than assumed (Sec. III-B); (ii) a ten-method benchmark against stated chance levels, with crop+context and crack-erased controls (Sec. V); (iii) a synthetic revisit protocol isolating appearance change from damage evolution, the only multi-visit evidence this paper has (Sec. VII-D); (iv) a geometric reference baseline and a protocol for this task’s failure modes: coverage-aware metrics, unanswerable queries, and an admission gate (Sec. VI, IV, VII).

II. RELATED WORK

Crack analysis is overwhelmingly focused on detection and segmentation [5]–[7]; its datasets annotate cracks within single images and the large consolidations merging [1], [2] inherit that scope, and multi-view crack pipelines fuse several photographs of one inspection into a surface or a segmentation [8] without solving instance correspondence between them. Structural-health monitoring does compare cracks across time with machinery close to ours [9]–[12], but identity is always *given by construction* (fixed camera, marked location, operator at a viewpoint): the subject is one known crack, the question of width or extent, and no dataset supports method comparison on the association decision itself. Those systems therefore solve measurement after correspondence has already been supplied; they do not decide whether a newly observed crack is the same physical defect as a gallery crack under an open-set protocol. Our benchmark targets that missing association decision, while its real capture scope remains within one walk-around rather than a true longitudinal study.

Object re-identification [4] with open-set identification [13] is the formal setting we adopt; keypoint matching [14], [15], learned correspondence (SuperGlue [16], LoFTR [17]), and

region embeddings [3], [18]–[21] apply directly to crack crops, and our benchmark measures all of them. Vehicle-damage assessment already re-identifies damage across claims in curated settings [22], a task our open-set protocol mirrors; the rest of that literature treats damage detection and severity rating rather than association [23]–[25]. Our results suggest the crop scope is itself the limiting choice. The move our baseline makes, aligning two captures and then reasoning about what is present in one and absent in the other, is change detection [26], [27]; planar registration is classical [28]–[30] with photometric alignment (ECC [31]) as the fallback. Our contribution is not a new estimator but the observation that the alignment target and the matching target should be *disjoint*: the homography is driven by the wall precisely because the crack is what we seek to identify.

III. CRACKID

A. Capture and composition

We photographed 17 exterior walls at one site over four half-days with two consumer handsets, yielding 140 photographs at 4080×3072 . The walls span two surface families that behave very differently under keypoint matching: stippled render (densely textured) and smooth painted plaster, where the crack is frequently the only structure in frame. Walls split disjointly into 6 validation walls (66 photographs, 66 identities) and 11 test walls (74 photographs, 146 identities). Statistical power is governed by identities visible in two or more photographs, since only these form a query with an answer: 18 in validation, 19 in test. Each wall was captured in one continuous walk-around (consecutive frames a median of 3 s apart), so the dataset measures **viewpoint and scale correspondence** within one visit, an upper bound on multi-visit performance, not yet a measurement of it.

The adjacent frame is always an answer. For 100% of the 280 answerable test queries, the nearest correct gallery entry is the adjacent frame. Ranking numbers are therefore near-duplicate retrieval; Sec. VII sweeps the exclusion gap to show the structure is not what produces the result.

B. Annotation, and why it needs auditing

One click point per crack is propagated across consecutive photographs through an ORB homography and one-to-one assignment; identities whose components appear to be fragments of one physical crack are then merged. The merge is where risk concentrates: a wrongly joined merge converts a hard negative into a free positive and inflates *every* method’s mAP together, so it cannot be caught by comparing methods. We therefore report its extent rather than assume its correctness: 95/212 **identities (45%) were formed by a merge** (48% on test), bridging a median maximum gap of 228 px; the adjusted Rand index against the pre-merge labeling is 0.080, meaning the merge restructured the partition rather than refining it. We release the instrumentation for an audited error rate (per-identity provenance, a stratified sample with contact sheets, Wilson scoring) rather than the audit itself; the verdicts are not yet filled in, so we claim no sampled error rate.

TABLE I
COUNTING UNITS FOR EVERY METRIC. PER SPLIT: WALLS,
PHOTOGRAPHS AND IDENTITIES; IDENTITIES IN TWO OR MORE
PHOTOGRAPHS (THE ONLY ONES THAT CAN FORM A QUERY WITH AN
ANSWER); ANSWERABLE QUERIES (RANKING POOL); POSITIVE AND
VALID QUERY-GALLERY PAIRS (PAIRWISE- F_1 POOL, TEST PREVALENCE
0.180); AND SAME-WALL IMAGE PAIRS (REGISTRATION POOL,
SEC. VII-B).

	Validation	Test
Walls / photographs / identities	6 / 66 / 66	11 / 74 / 146
Identities in ≥ 2 photographs	18	19
Answerable queries	439	280
Positive query-gallery pairs	26 012	5 722
Valid query-gallery pairs	89 118	31 822
Same-wall image pairs	394	301

IV. BENCHMARK PROTOCOL

A query is one annotated crack instance in one photograph; the gallery is the query’s own wall, so hard negatives share material and lighting; the query’s own photograph and week are excluded, leaving $n_Q = 280$ scored queries on test. A *query-gallery pair* is a query matched against one gallery instance of the same wall, and an *image pair* is two photographs of one wall. Table I gives these units per split. Registration (Sec. VI) is charged per image pair, but only the pairs containing at least one scored crack are actually registered, which on test is 255 of the 301; the remaining pairs are those whose photographs carry no crack instance the segmenter passes its size filter. Three choices deliberately favor no method, including ours:

- **Unanswerable queries are retained.** A query whose identity has no gallery counterpart can contribute false alarms and never successes, which is exactly the open-set case a maintenance system faces.
- **Thresholds are calibrated on validation.** Every method reports an oracle-thresholded F_1^* (generous to all) and a deployable $F_1^{\text{@v}}$ thresholded on held-out walls.
- **Missing scores are charged, not skipped.** A pair a method cannot score counts as predicted-negative and ranks last; coverage is not independent of the label (Sec. VII-B).

A. Metrics

We report Rank-1, Rank-5 and mAP, pairwise F_1 , and assignment F_1 (a one-to-one matching per image pair, the deployed operation). The headline metric is the open-set **detection-and-identification rate at 10% false alarm** (DIR@FAR=0.1): a known query is correct only if its top-ranked match is above threshold and correct; an unknown query is a false alarm whenever anything clears the threshold.

Every metric is reported against its chance level, because two are far from zero: pairwise F_1 has a best-constant-predictor floor $2p/(1+p) = 0.305$ on this pool, and Rank-5 chance is 0.767; Rank-1 and DIR@FAR chance are 0.336 and 0.038 (closed form, plus 200 random matrices through the same evaluation code). This is a prevalence-induced floor, not a coin-flip chance rate, and it changes when the evaluated

pair pool changes. Confidence resamples **walls**, not queries, since queries from one wall succeed or fail together, and with 11 test walls any such interval would be wide; Sec. VIII states why we report none.

V. THE APPEARANCE CEILING

A. Setup

Ten crop-scope matchers run under an identical protocol, identical inputs, and identical scoring: handcrafted keypoints (SIFT [14], ORB [15]), learned correspondence (SuperGlue [16], LoFTR [17]), general embeddings (ViT-B/16 [18], DeiT-S [19], CLIP ViT-B/32 [20], DINOv2 [21], YOLOv8 [32]), and a re-identification architecture (OSNet [3]).

B. Result: appearance buys almost nothing on the decision

On the decision a record actually makes, same crack or not, at a threshold, no crop method clears chance by more than 0.054 (excess over the 0.305 floor at the oracle threshold, Table II: 0.000 to 0.054, three methods exactly at the floor, nine within 0.052), across two decades of methods and more than two orders of compute (0.011s to 8.6s per query, Table II). At a deployable (validation-calibrated) threshold, nine of the ten span only -0.012 to $+0.012$ around chance; **LoFTR is the exception**, $F_1^{\text{@v}} = 0.351 (+0.046)$, and the most expensive crop method. Ranking does not transfer to deciding: the ten separate over a 0.302 range on Rank-1, yet OSNet ties for best Rank-1 (0.654) while its oracle pairwise F_1^* sits only 0.004 above the floor, and SuperGlue (fifth on Rank-1) has the best crop open-set rate (0.429). Ordering a gallery well and keeping a record are different capabilities. The collapse after validation calibration is consistent with threshold instability under a small, wall-clustered sample: methods can order candidates differently while their score distributions do not support a stable same-crack/not-same-crack boundary on held-out walls. This is an interpretation of the observed pattern, not a claim that calibration itself is the sole cause.

One contributor is checkable from Table I alone. Validation pair prevalence is 0.292 (26 012/89 118) against 0.180 on test, so the F_1 -optimal threshold is not a property of a method but of the pool it was fitted on: the same scores that separate cracks on the denser validation pool place their boundary in the wrong place once prevalence falls, and every method pays the same shift. The pairwise- F_1 floor moves with it, from 0.452 to 0.305. This is a mechanism, not a measurement: the threshold-free version of the claim (pairwise average precision, or AUROC against prevalence) would settle it, and we do not report it because the saved score matrices it requires are not released (Sec. VIII).

C. Is it the crack, or the wall behind it?

Two controls separate identity cue from field of view: a *crop+context* condition (wider crop) and a *crack-erased* condition (the crack inpainted out of that widened view). Context helps a lot: OSNet’s Rank-1 rises from 0.654 to 0.775, mAP to 0.595, open-set rate from 0.168 to 0.404, and its

TABLE II

MAIN RESULT, TEST SPLIT, $n_Q = 280$ QUERY INSTANCES OVER 11 WALLS. ALL METHODS SEE IDENTICAL INPUTS AND ARE SCORED BY IDENTICAL CODE. F_1^* IS ORACLE-THRESHOLDED ON TEST; ΔF_1^* IS ITS EXCESS OVER THE 0.305 BEST-CONSTANT-PREDICTOR FLOOR ($2p/(1+p)$), NOT COIN-FLIP CHANCE; SEC. IV); $F_1^{@v}$ IS THRESHOLDED ON THE HELD-OUT VALIDATION WALLS AND IS THE DEPLOYABLE NUMBER. *scored* IS THE FRACTION OF VALID QUERY–GALLERY PAIRS RECEIVING A FINITE SCORE; UNSCORED PAIRS ARE CHARGED AS PREDICTED-NEGATIVE, AND THE QUERY IS STILL COUNTED (SEC. VII-B). COST IS WALL-CLOCK SECONDS PER QUERY. THE COVERAGE-ONLY REFERENCE ROW ANSWERS YES TO EVERY REGISTERED PAIR WITHOUT EXAMINING A CRACK AND IS THE LEVEL EVERY OTHER ROW MUST BE READ AGAINST; CHANCE LEVELS FOR EVERY COLUMN ARE GIVEN IN SEC. IV. BEST CROP-SCOPE VALUE PER COLUMN IN **BOLD**.

Method	Family	R@1	mAP	DIR@FAR=0.1	F_1^*	ΔF_1^*	$F_1^{@v}$	scored	s/query
<i>Reference level</i>									
COVERAGE-ONLY	geometry	0.515	0.533	0.000	0.544	0.239	0.544	0.48	—
<i>Crop scope</i>									
ORB [15]	keypoint	0.352	0.364	0.018	0.305	0.000	0.305	1.00	0.038
SIFT [14]	keypoint	0.432	0.417	0.036	0.313	0.008	0.305	1.00	0.086
SuperGlue [16]	learned	0.562	0.455	0.429	0.353	0.048	0.305	1.00	5.826
LoFTR [17]	learned	0.549	0.440	0.336	0.359	0.054	0.351	1.00	8.639
YOLOv8 backbone [32]	embedding	0.561	0.453	0.157	0.357	0.052	0.308	1.00	0.217
ViT-B/16 [18]	embedding	0.654	0.511	0.286	0.331	0.026	0.317	1.00	0.193
DeiT-S [19]	embedding	0.621	0.497	0.250	0.305	0.000	0.293	1.00	0.058
CLIP ViT-B/32 [20]	embedding	0.604	0.471	0.143	0.307	0.002	0.306	1.00	0.050
DINOv2 [21]	embedding	0.607	0.484	0.307	0.305	0.000	0.297	1.00	0.011
OSNet [3]	re-ID	0.654	0.520	0.168	0.309	0.004	0.309	1.00	0.053
<i>Full-image scope</i>									
REG+CHAMFER (reference)	geometry	0.952	0.886	0.768	0.656	0.351	0.582	0.48	27.92

pairwise F_1^* from 0.309 to 0.354; CLIP’s open-set rate rises from 0.143 to 0.311, **but erasing the crack costs almost none of it**: OSNet retains Rank-1 0.711, mAP 0.574 and open-set 0.393; CLIP retains Rank-1 0.600, mAP 0.502, open-set 0.286. Both erased views still outrank every tight-crop method: the retrieved cue is the wall, that is, location. **Pairwise F_1 moves little with context**: all four context variants lie between 0.346 and 0.361, at most 0.056 above the floor, over a change that gives OSNet +0.236 of open-set rate. We read the aggregate as a property of the data: methods with no shared inductive bias land near the floor, and wherever a crop method improves, the improvement is not the crack. Ten methods are a sample, not proof; the result establishes that the obvious moves buy 0.000 to 0.054 above chance on the decision, reason to change the formulation, not the encoder.

VI. A GEOMETRIC REFERENCE BASELINE

The ceiling motivates a change of scope. A crack is a mark at a fixed location on a rigid, near-planar surface that carries texture the crack does not; we identify a crack by *where it is on its wall* once the two photographs are aligned. We release this as a deliberately simple *reference baseline* for CRACKID, with no learned component beyond the shared crack segmenter.

A. Alignment driven by non-crack texture

A wall is planar, so the homography $x' \sim Hx$ mapping photograph I_a into I_b ’s frame is the correct model. The design decision that matters is **what drives the fit**: we dilate the crack masks by 15px and exclude those pixels from detection, so H comes from wall texture alone. This is not an optimization but what makes the projected position an independent identity cue: were cracks to participate, a long crack could dominate

the correspondence set and pull the fit toward the wrong crack. Detection runs on copies downsampled to 1600px under CLAHE equalisation; we use SIFT (up to 8000 features), a 0.75 ratio test, and MAGSAC at 4px requiring 25 inliers.

B. A cascade for two surface families

One front end does not serve both surfaces, and the failure is categorical: On the smooth-plaster walls, crack-masked SIFT succeeds on *exactly zero* pairs. We try three in order: (i) **SIFT on masked wall texture**, for stippled render; (ii) **SIFT with cracks restored**, triggered only when masking starves detection below 300 keypoints (on plaster the crack is the only feature in frame), accepting the circularity stage (i) avoids, and recording which front end produced each result; (iii) **ECC**, direct photometric alignment over a $320 \rightarrow 640 \rightarrow 1280$ px pyramid, on the large-scale shading that survives on flat paint. Every candidate H is double-checked (reject mirrored or folded quadrilaterals, area scale outside $[1/16, 16]$, collapsed sides); its inlier RMS sets downstream tolerances.

C. Projection and Chamfer agreement

Crack instances are connected components; pixels from I_a are projected by H and compared with instances of I_b . Chamfer distance, not IoU, is the criterion: cracks are a few pixels wide, so a 3px registration error drives IoU to zero for a correctly aligned pair. The reported score is the **symmetric mean Chamfer distance** in pixels

$$D(a, b) = \frac{1}{2} \left[\frac{1}{|a|} \sum_{p \in a} d(p, b) + \frac{1}{|b|} \sum_{q \in b} d(q, a) \right], \quad (1)$$

over the union bounding box, with cheap rejection when boxes are far apart. We also report a scale-free alternative, **Chamfer**

coverage, the symmetric fraction of pixels within τ , because Eq. (1) is a per-pair pixel distance whose scale differs per pair, so a single global threshold makes ranking partly measure calibration (Sec. VII-C).

D. Explicit non-answers, and the licensed negative

Correspondence is a one-to-one assignment per image pair with an explicit rejection threshold, not a per-query argmax: two cracks must not both claim the same crack; unmatched instances are reported as *disappeared* or *newly appeared*. The Hungarian solve is part of *evaluation*, applied identically to every method’s score matrix. Specific to this formulation is the evidence a **licensed negative** can carry: with I_a projected into I_b ’s frame, the system observes a region visible and empty of the recorded crack, positive evidence of absence, categorically stronger than the low similarity a crop embedding produces. A projected crack falling outside the gallery photograph is a confident rejection, counted separately from unscorable pairs; this is the mechanism behind the open-set gap (Sec. VII-B).

VII. RESULTS

A. Main comparison and the open-set gap

Relocating identity into the wall raises every metric: Rank-1 0.654 to 0.952, mAP 0.520 to 0.886, and as excess over the COVERAGE-ONLY reference the ranking gains are $+0.437/+0.353$, while pairwise $F_1^* = 0.656$ is $+0.112$ over 0.544, not the $+0.351$ a comparison against the crop floor would suggest. The **open-set result is the one number coverage does not explain**: DIR@FAR=0.1 reaches 0.768 against 0.429 for the best crop method (SuperGlue), 0.404 for the best context control, 0.038 for chance, and exactly **0.000** for COVERAGE-ONLY. Crop methods cannot decline: they return a similarity to the nearest neighbour, which on self-similar hairline cracks is often high, while an aligned pair supports the *licensed negative*: the query projects into the gallery frame, where its region is visible and empty. Compute does not explain the gap: SuperGlue’s 0.429 costs 5.8 s/query, OSNet’s 0.168 costs 0.053 s.

B. Partial coverage, gap structure, and failures

Every crop method scores 100% of valid pairs, the geometric method 48%; Unscorable pairs are charged as predicted negatives. For **ranking** that is a penalty: censoring a complete matrix drops Rank-1 0.422 to 0.27 at 34% coverage. For **pairwise** F_1 it is a subsidy: the scorable subset retains 100% of positives and 36.8% of negatives, so prevalence rises 0.180 to 0.373 and the floor with it, 0.305 to 0.544. COVERAGE-ONLY scores 0.544 pairwise F_1 and mAP 0.533, both above OSNet (0.309 and 0.520), and Rank-1 0.515, above ORB and SIFT but below the embedding and learned crop methods, by answering every registered pair without examining a crack. Its signal is registration overlap: it assigns one constant score to registered image pairs and $-\infty$ to unregistered pairs, so candidate images are ranked even though crack pixels are never inspected. Much of the ranking and pairwise- F_1 margin

is therefore *which pairs align*; the open-set metric is the exception and the result we lead with.

Excluding the adjacent frame, then the two and three nearest, leaves Rank-1 flat (OSNet 0.654 to 0.648) with excess over chance stable in 0.24 to 0.32: the near-duplicate structure is real but not what produces the numbers. It is also important for interpretation: the adjacent frame answers all 280 queries, whereas the gap-3 sweep leaves only 105 answerable queries, so the headline ranking measures short-baseline correspondence rather than broad temporal re-identification. Registration success is strongly wall-dependent (97/301 pairs, 32.2%: walls 07/08/12 100%, wall 09 52%, wall 11 83%, walls 13-16 between 3% and 29%, walls 10/17 at 0%). The pairs that register are the densely annotated ones, so the 96 registered pairs that carry a scored query-gallery pair hold 48% of the 31 822 valid ones — 160 valid cells per image pair against 104 for a failed pair — from a third of the available pairs (46 of the 301 same-wall pairs yield no valid pair at all). A logistic fit over all 301 pairs makes the mechanism plain: sharpness dominates (median 133.8 registered vs 7.2 failed, wall-clustered Wald $z = 5.1$, $p = 3 \times 10^{-7}$), frame gap far less ($z = -2.8$, $p = 0.005$). The independent-pairs z s (8.2, -5.0) overstate the contrast because failure is categorical by wall, so we resample walls, not queries, and the leave-one-wall-out spread in Rank-1 is large, so no fine ranking among baselines is claimed. One admission gate (variance-of-Laplacian ≥ 10) applied identically rejects 21 of 74 test captures and nearly doubles registration, 32.2% to 56.1%.

C. Ablations and cost

Chamfer distance versus coverage. The scale-free coverage score motivated in Sec. VI is not a free improvement: it drops mAP 0.886 to 0.646 and Rank-1 0.952 to 0.849, the largest movement any change produces; DIR and $F_1^{@v}$ barely move, so coverage gives a threshold decision, not a ranking. **Masking.** Disabling crack exclusion leaves ranking and pairwise agreement practically unchanged (to three decimals) but raises DIR@FAR 0.768 to 0.825: masking is not what buys the reported accuracy. This is expected when the test set contains no deliberate case where crack pixels create a spurious registration: masking buys evidence independence, not observed accuracy on this collection, and should be treated as a safeguard awaiting a harder stress test. **ECC.** Disabling ECC (repaired after a pipeline bug silently left it inert) gives Rank-1 0.920, mAP 0.862, coverage 0.50 against 0.952, 0.886, 0.48; assignment F_1 rises 0.769 to 0.784 when enabled; open-set is unchanged.

Cost. At 27.9 s/query the baseline is more than two orders of magnitude above OSNet, but that per-query figure is a charging convention, not an intrinsic one: registration is per image *pair*, one homography serves every crack on a wall, and amortized over the 301 test image pairs the same pass runs about 26 s/pair (7817 s total), after which any later crack query on the pair pays only the Chamfer pass, not the homography. That pair-level cost is within an order of magnitude of LoFTR (8.3 s), a crop method with none of the open-set benefit.

Protocol	Method	n_q	R@1	mAP	DIR@FAR=0.1	F_1^{div}	F_1^a	scored
A (growth)	Hybrid	69	0.275	0.401	0.343	0.569	0.646	1.00
A (growth)	OSNet+ctx	69	0.826	0.647	0.739	0.601	0.631	1.00
B (appearance)	Hybrid	82	0.274	0.403	0.342	0.574	0.640	1.00
B (appearance)	OSNet+ctx	82	0.671	0.570	0.663	0.594	0.609	1.00

TABLE III

HYBRID VERSUS THE STRONGEST APPEARANCE BASELINE ON THE SYNTHETIC REVISIT PROTOCOL (SEC. VII-D); PAIRWISE AND ASSIGNMENT F_1 AT THE VALIDATION-CALIBRATED THRESHOLD. ONE SEED (0), TEN SOURCE PHOTOGRAPHS ACROSS FIVE WALLS, NO RESAMPLING; ROWS CARRY NO UNCERTAINTY AND ARE AN ACCURACY CHECKPOINT, NOT A STABILITY CLAIM (SEC. VII-D).

D. A synthetic revisit protocol

The maintenance setting these records serve is a *revisited* crack, so we add a second protocol isolating controlled change. Ten source photographs across five of the walls (walls 01, 02, 04, 05, 09; one seed, 0) are each synthesized into a set of queries under an identical warp ($1.08\times$ scale, 6° rotation, 4° tilt): dataset A additionally grows every crack tip (damage evolution), dataset B corrupts appearance while leaving structure untouched (blur $\sigma=1.6$, noise $\sigma=14$, brightness/contrast shift); each synthetic query is ranked against the wall’s real instances only. Unlike the Table II REG+CHAMFER system, the synthetic-protocol *Hybrid* folds a structural skeleton comparison into the geometric baseline as a fallback plus per-pair blend, scored by the same code as OSNet+ctx (Table III).

The geometric side does not transfer its main-benchmark advantage here: OSNet+ctx separates the synthetic queries at R@1 0.826/0.671, the hybrid at 0.275/0.274, with open-set rates 0.739/0.663 against 0.343/0.342. But the split is metric-dependent, and opposite the ranking gap: on identity agreement, the hybrid holds its own, with assignment $F_1^a = 0.646/0.640$ beating OSNet’s 0.631/0.609, and pairwise $F_1^{div} = 0.569/0.574$ trailing 0.601/0.594 by far less than the ranking margin. Both rate 1.00 scored, and per cell none was decided by the pure-homography branch alone ($H\% = 0.0$, the share of assignment cells whose verdict came straight from the homography, with no blend or skeleton fallback): the geometric pipeline answers *which crack is which* on pairs appearance also resolves, but loses the global ranking of a hundred-plus same-wall candidates at one threshold. These rows are a checkpoint, not a claim: one seed, one wall subset.

VIII. THREATS TO VALIDITY

We state the limits in the body because they bound the central claims.

The effective sample is 19 identities. The test split’s 280 queries come from 19 identities visible in two or more photographs across 11 walls; queries from one wall share material, illumination, and one camera pass, so query count is not sample size. We resample walls and report the leave-one-wall-out spread in Rank-1. The same resampling discipline applies to the open-set headline: DIR@FAR=0.1 of 0.768 is a point estimate at one operating point on the same sample, so the abstract compares it to baselines as a one-number

structure, an open-set gap of roughly 0.34 to 0.36 over the best crop and best context methods, and claims nothing about the precise spread between resampled walls. This benchmark supports a structural claim about the formulation, but not fine-grained ranking among baselines. We do not attach a numeric wall-resampled interval to the abstract point estimates because the score matrices needed to recompute it are not part of the released artifacts; the effective sample and wall-level bimodality should therefore be read as a limitation, not as hidden precision.

Scope is controlled, and the control found something.

The method reads full photographs; the baselines read crops, dictated by design, since homography needs wall context. Two controls (Sec. V-C) separate identity cue from field of view, and scope does not account for the open-set gap (0.404 best context against 0.768). But the crack-erased control is uncomfortable: inpainting the crack costs only 0.011 of that 0.404, so a context-scope number is partly background-matching, which we expect to weaken on real revisits. Ranking numbers are additionally near-duplicate (the adjacent frame answers all 280 queries, Sec. III); the gap sweep shows the trend is flat, with power falling to 105 answerable queries by gap 3.

The annotation and baseline are circularly coupled.

Identities were seeded by propagating one click point between consecutive photographs through an ORB homography (Sec. III-B), and the geometric baseline’s whole mechanism is homography registration. Of the 5 722 positive (same-crack) query-gallery pairs on test (Table I), 5 718 (99.9%) involve an identity formed by a merge; only 4 pairs, spanning 2 identities, join two identities that were never merged. Thus the clean subgroup needed to test whether geometry helps independently of annotation provenance does not exist at useful power. The two reported checks are limited: the procedures are not identical, and merged versus unmerged positives register at the same observed rate (100% both ways: 83/83 and 1/1 of the image pairs that carry one), but the latter cannot test whether a biased homography would register differently from an unbiased one, because both alignments are related procedures. Nor did we run a sensitivity analysis quantifying how much false-positive mass or registration selection would be required to manufacture an open-set gap of that size; the saved score matrices needed for that analysis are not released. Accordingly, the geometric advantage remains unresolved with respect to circularity, rather than being demonstrated independent of it. The decisive follow-up is a small, manually verified identity subset labeled without homography propagation, reported even if underpowered, together with a bias-sensitivity curve.

Annotations are instrumented, not audited. The human verdicts that would turn the provenance of Sec. III-B into a sampled error rate with a confidence interval have not been collected; the 45% merge rate is reported, and relative comparison is robust to it since the merge policy hits all methods identically. All captures come from one site over four half-days, two handsets and two surface families; smooth plaster produces the categorical SIFT failure noted

in Sec. VII-B; cracks are located by a learned segmenter whose errors propagate into Chamfer agreement unquantified. What would convert the within-visit benchmark into a cross-visit measurement is a real second capture session at marked distances and angles; we have not run one. The synthetic revisit protocol is therefore a controlled check, not a substitute for multi-visit data, and the paper does not claim cross-visit performance.

IX. CONCLUSION

We built a benchmark that makes crack re-identification testable: CRACKID, 140 photographs over 17 walls with 212 identities, its annotation provenance (45% merged) stated rather than assumed. The benchmark’s main finding is negative and structural: on the decision a record requires, ten matchers across four families and more than two orders of compute buy between 0.000 and 0.054 over chance, and only LoFTR survives a validation-calibrated threshold. Ordering a gallery and deciding an identity are not the same capability. The difficulty is not crack appearance: a geometric reference baseline that registers on wall texture with cracks masked out and Chamfer-matches projected masks reaches $\text{DIR@FAR}=0.1$ of 0.768 against 0.404 for the best context method while answering 48% of pairs, and a control exploiting that alignment coverage without examining a crack scores **0.000**. Operationally, the open-set advantage comes from an aligned pair supporting a *licensed negative*, positive evidence of absence, where a crop embedding can only report low similarity; whether that advantage is independent of homography-based annotation remains unresolved. Two bounds remain addressable: a real second capture session turns the benchmark from viewpoint to cross-visit, and the annotation audit awaits its verdict. The 48% coverage is also a deployment gap in its own right, not merely a scoring issue: smooth-plaster walls can be categorically unanswerable. Finally, the synthetic revisit protocol reverses the headline ranking and open-set result in favour of OSNet+context, while the geometric pipeline retains only a narrow assignment- F_1 advantage; the present paper therefore establishes a promising within-visit geometric reference, not a solved multi-visit maintenance system.

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